Solar ESS Co. Ltd.
53 Moo 2 Tambon Nanglae, Amphoe Muang, Chiang Rai 57100
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TECHNICAL NOTE · BATTERY COMPARISON

Three 16 kWh low-voltage batteries, side by side

Solis IntelliHome, Dyness PowerBrick Plus and Pylontech Fidus Battery Plus, compared for an off-grid homestay on a Solis 16 kW hybrid inverter.

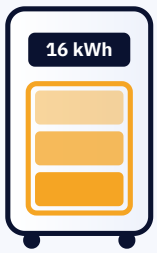


Solis IntelliHome

IntelliHome-16kWh-OD · 51.2 V LiFePO4

14.40 kWh usable

6,000 cycles · IP66

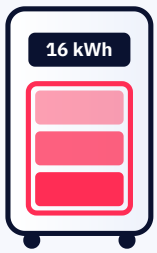


Dyness PowerBrick Plus

PowerBrick Plus 16.07 kWh · 51.2 V LiFePO4

15.27 kWh usable

8,000 cycles · IP65



Pylontech Fidus Plus

FB-L-16 · 51.2 V LiFePO4

15.28 kWh usable

8,000 cycles · IP65

Specification at a glance

	Solis IntelliHome	Dyness PowerBrick Plus	Pylontech Fidus Plus
Nominal energy	16.00 kWh	16.07 kWh	16.08 kWh
Usable energy	14.40 kWh	15.27 kWh	15.28 kWh (16.08 at 100%)
Current per unit	157 A	200 A (300 A for 2 min)	200 A datasheet; 0.5C (≈157 A) continuous under warranty
Cycle life	6,000	8,000	**8,000 · 15-yr design life**
Outdoor rating	**IP66**	IP65	IP65 + C5-M anti-corrosion
Fire suppression	Not stated	**Built in**	Aerosol (listed as optional)
Max in parallel	8 units	**50 units**	20 units per group
Size · weight	470 × 950 × 240 mm · 126 kg	**435 × 235 × 850 mm · 119 kg**	435 × 240 × 900 mm · 130 kg
Pairing with Solis S6	**Same brand as the Solis inverter**	Solis listed as compatible by Dyness	To confirm on the Solis approved list

Green = strongest in that row. Specifications from each maker's published datasheet.

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WHAT THE NUMBERS MEAN

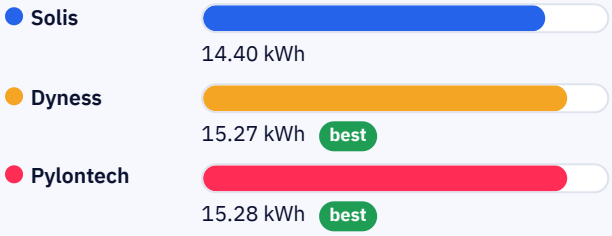
Energy, lifetime and installation

Longer bars are better, except for weight, where shorter is better.

Capacity

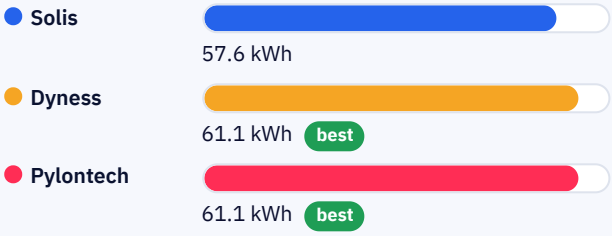
Usable energy per battery

What you can actually take out of one unit.



Usable energy with 4 batteries

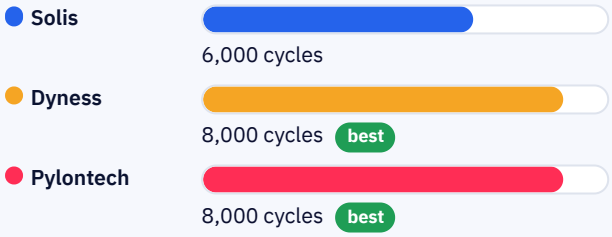
The size proposed for the homestay.



Lifetime

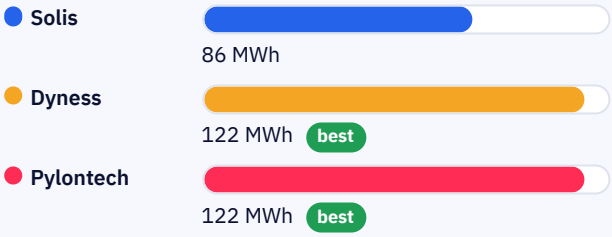
Rated cycle life

Full charge–discharge cycles in the maker's test.



Energy over the battery's life

Usable energy × rated cycles, per battery, on paper.



Installation

Weight per usable kWh

Lower is lighter to move and mount.



Room to grow

Most units the maker allows in parallel (Pylontech: per group).



Lifetime figures are calculated from rated cycles at 25 °C and are a guide to compare brands, not a guarantee. Pylontech states its warranty as energy delivered: 2.8 MWh per kWh, about 45 MWh per battery, and limits continuous use to 0.5C. Real life depends on temperature, depth of discharge and charge rate.

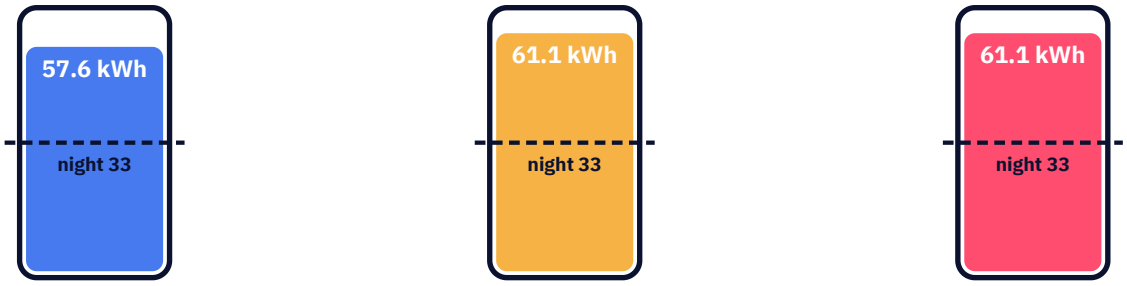
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PROJECT FIT · OFF-GRID HOMESTAY

What it means for a 4-battery system

16 kW	≈ 33 kWh	≈ 80 A	4 × 16 kWh
Solis S6 hybrid inverter, 1 phase	night use from the battery	per battery at full inverter output	batteries in parallel

Energy left after one night



+25 kWh spare	+28 kWh spare	+28 kWh spare
Solis × 4	**Dyness** × 4	**Pylontech** × 4

All three cover one night with room to spare for cloudy days. With 4 units sharing the load, each battery carries about 80 A, well inside every brand's continuous limit, so the current difference does not matter here.

Which one to choose

Solis IntelliHome

Simplest · one brand

- Same brand as the inverter: one warranty contact
- Native pairing, no approval step
- Fewer rated cycles (6,000)

Dyness PowerBrick Plus

More energy · longer life

- About 6% more usable energy than IntelliHome
- 8,000 cycles and built-in fire suppression
- Up to 50 units in parallel

Pylontech Fidus Plus

Strongest on paper · confirm Solis approval

- Most usable energy, C5-M anti-corrosion
- 8,000 cycles, 15-year design life
- Must be on the Solis S6 approved list

What to check

Inverter approval
Check each battery on the Solis S6 approved battery list.

Size for cloudy days
Keep spare energy after the night for days with little sun.

Parallel parts
Confirm the bus bar or hub needed for 4 units of each brand.

Warranty terms
Compare years and energy-delivered limits in writing.

In short: all three cover an off-grid night with energy to spare. IntelliHome is the simplest one-brand choice for a Solis system. Dyness and Pylontech give more energy and longer rated life; check Solis approval first. Ask us for a price for your home.